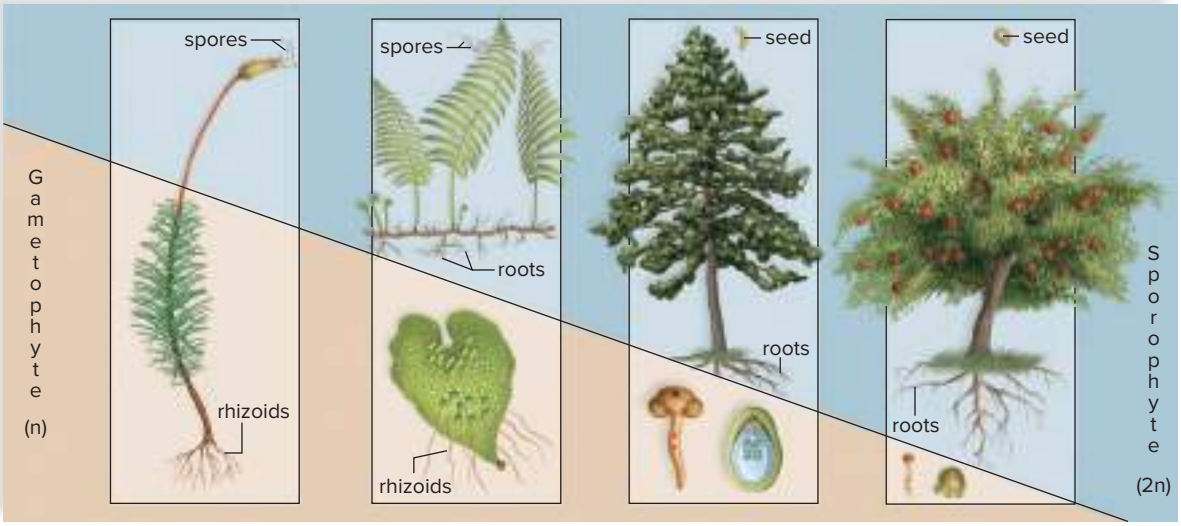


Figure 23.6 Reduction in the size of the gametophyte. Notice the reduction in the size of the gametophyte and the increase in the size of the sporophyte among these representatives of today's land plants. This trend occurred as these plants became adapted for life on land. In the moss and fern, spores disperse the gametophyte. In gymnosperms and angiosperms, seeds disperse the sporophyte.

Check Your Progress 23.1

1. Compare and contrast the traits of charophytes and land plants.
2. List the adaptations that led to a land existence for plants.
3. Identify the role of each generation in the alternation-of-generations life cycle.

23.2 Evolution of Bryophytes: Colonization of Land

Learning Outcomes

Upon completion of this section, you should be able to

1. List the traits that classify a plant as a bryophyte.
2. Compare the three groups of bryophytes.
3. Identify the key structures and stages in the life cycle of a moss.

The **bryophytes**—liverworts, hornworts (Anglo-Saxon *wort,* "herb"), and mosses—were the first plants to colonize land. They only superficially appear to have roots, stems, and leaves, because, by definition, true roots, stems, and leaves must contain vascular tissue. **Vascular tissue** is specialized for the transport of water and organic nutrients throughout the body of a plant. Bryophytes, which lack vascular tissue, are often called **nonvascular plants.** Vascular tissue also provides support to the plant body, so bryophytes typically are low-lying; some mosses reach a maximum height of only about 20 cm.

The fossil record contains some evidence that the various bryophytes evolved during the Ordovician period (about 443.7 MYA, see Table 18.1). An incomplete fossil record makes it difficult to tell how closely related the various bryophytes are. Molecular data, in particular, suggest these plants have individual lines of descent, as shown in Figure 23.1, and that they do not form a monophyletic group. The observation that today's mosses have a rudimentary form of vascular tissue suggests they are more closely related to vascular plants than to hornworts and liverworts.

Bryophyte bodies are covered by a cuticle, which is interrupted in hornworts and mosses by stomata, and they have apical tissue that produces complex tissues. However, bryophytes are the only land plants in which the gametophyte is dominant (see Fig. 23.6). Antheridia produce flagellated sperm, which means they need a film of moisture in order to swim to eggs located inside archegonia. The bryophytes' lack of vascular tissue and the presence of flagellated sperm mean you are apt to find bryophytes in moist locations. Some bryophytes compete well in harsh environments because they can reproduce asexually.

Bryophytes contribute to the lush beauty of forests. They can convert mountain rocks to soil, hold moisture and metals, and resist desiccation. Scientists working in genetic engineering are interested in bryophytes' ability to resist chemical reagents, decay, and herbivory. The idea is that these traits could be transferred to other plants.

Liverworts

Liverworts are divided into two groups—the thallose liverworts with flattened bodies, known as a thallus; and the leafy liverworts, which superficially resemble mosses. The name *liverwort* refers to the lobes of the thallus, which to some resemble the lobes of the liver. The majority of liverwort species are the leafy types.

The liverworts in the genus *Marchantia* have a thin thallus, about 30 cells thick in the center. Each branched lobe of the thallus is approximately a centimeter in length; the upper surface is divided into diamond-shaped segments with a small pore, and the lower surface bears numerous hairlike extensions called **rhizoids** (Gk. *rhizion,* dim. of "root") that project into the soil (Fig. 23.7). Rhizoids serve as anchors and assist in the absorption of water.

Marchantia species reproduce both asexually and sexually. Gemmae cups on the upper surface of the thallus contain gemmae,

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